

Upper Limb Osteology, Surface And Clinical Anatomy

MBS 300 - Courtesy of Dr. Evans

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I. Surface Anatomy of the Upper Limb

The ability to accurately identify and palpate landmarks is required for physical examination, locating sites for injections, and planning surgical incisions. Key palpable bony landmarks, muscle contours, and the superficial courses of some neurovascular structures serve as guides.

For instance, in the **shoulder region**, the entire length of the **clavicle** is palpable, from the **suprasternal (jugular) notch** medially, across the **sternoclavicular joint**, to the **acromioclavicular joint** laterally. The **acromion process** forms the bony shelf lateral to the acromioclavicular joint, and its posterior aspect continues as the **spine of the scapula**. The **coracoid process** can be palpated in the **infraclavicular fossa**. While the humeral head itself is deep, the **greater and lesser tubercles** of the humerus are palpable through the deltoid muscle inferior to the acromion. The **deltoid muscle** itself forms the characteristic contour of the shoulder and its three parts (anterior, middle, posterior) become prominent with specific arm movements (flexion, abduction, extension, respectively). The **posterior cervical triangle**, bounded by the clavicle, sternocleidomastoid, and trapezius, is an important landmark where the subclavian artery and parts of the brachial plexus may be palpated.

Moving to the **arm and elbow**, the **medial and lateral epicondyles** of the humerus are prominent bony projections at the elbow. The **olecranon process** of the ulna, the "point" of the elbow, is easily identified, especially in flexion. These three landmarks—medial epicondyle, lateral epicondyle, and olecranon—form an equilateral triangle when the elbow is flexed to 90 degrees and a straight line when extended, a relationship crucial for assessing elbow joint integrity. The **biceps brachii** and **triceps brachii** muscles form the major contours of the anterior and posterior arm, respectively. The **brachial artery pulse** can be palpated in the cubital fossa, medial to the biceps tendon.

In the **forearm and wrist**, the **shaft of the ulna** is subcutaneous and palpable along its entire posterior border, from the olecranon to the **ulnar styloid process** at the wrist. The **radial styloid process** is palpable on the lateral aspect of the wrist. The **head of the ulna** is visible and palpable as a prominence on the medial-dorsal aspect of the wrist, especially during pronation. Distally, Lister's tubercle on the dorsal radius serves as a pulley for the extensor pollicis longus tendon. Several tendons become prominent at the wrist with movement, such as those of the flexor carpi radialis, palmaris longus (if present), and flexor carpi ulnaris. The **radial artery pulse** is readily palpated lateral to the flexor carpi radialis tendon at the wrist.

The **hand** presents numerous palpable landmarks, including the **metacarpal heads** (knuckles) and **phalanges**. The **anatomical snuffbox**, a triangular depression on the dorsolateral aspect of the wrist at the base of the thumb, is bounded by the tendons of extensor pollicis longus medially and extensor pollicis brevis and abductor pollicis longus laterally. The floor of the snuffbox is formed by the scaphoid and trapezium bones, and the radial artery courses through it.

The interplay between these palpable bony points, the changing contours of muscles during contraction, and the known pathways of major nerves and vessels forms a topographical map essential for clinical practice. Deviations from normal surface anatomy, such as swelling, deformity, or abnormal muscle wasting, can indicate underlying pathology. Furthermore, knowledge of these relationships guides safe access for procedures like venipuncture in the cubital fossa or injections into the subacromial space.

II. Osteology of the Upper Limb

The skeletal framework of the upper limb provides support, facilitates movement through its articulations, and serves as attachment sites for muscles. It comprises the pectoral girdle (clavicle and scapula) and the bones of the free limb (humerus, radius, ulna, carpals, metacarpals, and phalanges).

A. Clavicle

The clavicle, or collarbone, is an S-shaped long bone that acts as a strut connecting the upper limb to the axial skeleton. Medially, its **sternal end** articulates with the manubrium of the sternum at the **sternoclavicular joint**. Laterally, its flattened **acromial end** articulates with the acromion of the scapula at the **acromioclavicular joint**. Key features include the **conoid tubercle** and **trapezoid line** on its inferior surface for attachment of the coracoclavicular ligaments, which are vital for shoulder stability. The clavicle is the first bone to begin ossification during embryonic development. Its subcutaneous position makes it vulnerable to fracture, often occurring at the junction of its middle and lateral thirds. Such fractures can, due to the clavicle's proximity to vital structures, pose a risk to the underlying subclavian vessels and brachial plexus. This highlights that the clavicle is more than a simple strut; it is a biomechanically crucial element whose integrity is vital for normal shoulder function and whose disruption can have significant neurovascular consequences.

B. Scapula

The scapula, or shoulder blade, is a large, flat, triangular bone located on the posterolateral aspect of the thorax. It has a body, a prominent spine that divides the posterior surface into the supraspinous and infraspinous fossae, and three borders (superior, medial/vertebral, and lateral/axillary) and angles (superior, inferior, lateral). The spine extends laterally to form the **acromion**, which articulates with the clavicle. The **glenoid cavity (fossa)**, located at the lateral angle, articulates with the head of the humerus to form the glenohumeral joint. The **coracoid process** is a hook-like projection anterosuperior

to the glenoid cavity, providing attachment for muscles and ligaments. The scapula's movements (elevation, depression, protraction, retraction, upward and downward rotation) significantly contribute to the overall mobility of the upper limb.

C. Humerus

The humerus is the single bone of the arm.

Proximal End: Features the smooth, rounded **head** (articulates with glenoid cavity), **anatomical neck** (margin of the head), **greater and lesser tubercles** (muscle attachments for rotator cuff muscles), and the **intertubercular (bicipital) groove** (passage for biceps tendon). The **surgical neck**, a constriction below the tubercles, is a common site for fractures and is closely related to the axillary nerve and posterior circumflex humeral artery.

Shaft: Has the **deltoid tuberosity** laterally for deltoid muscle attachment and the **radial (spiral) groove** posteriorly, which accommodates the radial nerve and profunda brachii artery. Fractures of the humeral shaft can thus endanger the radial nerve.

Distal End: Flattened and features the **medial and lateral epicondyles** (muscle attachments), the **trochlea** (articulates with ulna), and the **capitulum** (articulates with radius). Anteriorly, the **coronoid fossa** (receives coronoid process of ulna during flexion) and **radial fossa** (receives head of radius during flexion) are present. Posteriorly, the **olecranon fossa** accommodates the olecranon process of the ulna during extension. Supracondylar fractures are common in children and can compromise the brachial artery and median nerve.

D. Radius and Ulna

These are the two bones of the forearm, articulating with the humerus proximally and the carpal bones distally, and with each other at the proximal and distal radioulnar joints, allowing pronation and supination.

Ulna: Located medially. Proximally, it features the **olecranon process** (forms point of elbow), **coronoid process**, and **trochlear notch** (articulates with trochlea of humerus). The **radial notch** on the lateral side of the coronoid process articulates with the head of the radius. Distally, it has a small, rounded **head** and a **styloid process**.

Radius: Located laterally. Proximally, it has a disc-shaped **head** (articulates with capitulum of humerus and radial notch of ulna), a **neck**, and the **radial (bicipital) tuberosity** (attachment for biceps brachii). The shaft expands distally

to form a broad articular surface for the carpal bones and the prominent **styloid process** laterally. The distal radius also has an **ulnar notch** medially for articulation with the head of the ulna. Colles' fracture, a common fracture of the distal radius with dorsal displacement, often results from a fall on an outstretched hand (FOOSH).

E. Carpal Bones

The wrist contains eight small carpal bones arranged in two rows of four:

Proximal Row (lateral to medial): Scaphoid, Lunate, Triquetrum, Pisiform (a sesamoid bone within the flexor carpi ulnaris tendon).

Distal Row (lateral to medial): Trapezium, Trapezoid, Capitate, Hamate (with its characteristic hook).

These bones form the carpal arch, which is concave anteriorly and bridged by the flexor retinaculum, forming the carpal tunnel. The scaphoid is the most frequently fractured carpal bone, typically from a FOOSH. Its clinical significance is magnified by its tenuous blood supply, which is predominantly retrograde from branches of the radial artery entering its distal pole. Fractures, especially of the waist or proximal pole, can disrupt this blood supply, leading to a high risk of avascular necrosis and non-union. This makes the scaphoid a small bone with potentially large clinical problems, necessitating careful diagnosis and management.

F. Metacarpals and Phalanges

The palm of the hand is formed by five metacarpal bones, numbered I to V from lateral (thumb) to medial. Each metacarpal has a base (proximally, articulating with carpals and adjacent metacarpals), a shaft, and a head (distally, articulating with proximal phalanges to form the metacarpophalangeal joints or knuckles). The phalanges are the bones of the digits. The thumb (pollex) has two phalanges (proximal and distal), while the other four fingers each have three (proximal, middle, and distal). Each phalanx has a base, shaft, and head.

III. Clinically Significant Regions and Structures

Certain anatomical regions in the upper limb are of particular clinical importance due to the density of vital structures they contain or their susceptibility to injury and pathology.

A. Axilla

The axilla, or armpit, is a pyramidal-shaped space between the upper arm and the thoracic wall. Its boundaries are formed by muscles: anteriorly by pectoralis major and minor; posteriorly by subscapularis, teres major, and latissimus dorsi; medially by the serratus anterior and upper ribs; and laterally by the intertubercular groove of the humerus. The apex is directed towards the root of the neck, and the base is formed by skin and axillary fascia. The axilla serves as a major conduit for neurovascular structures passing from the neck and thorax to the upper limb. Its contents include the axillary artery and its branches, the axillary vein and its tributaries, the cords and branches of the brachial plexus, and numerous axillary lymph nodes embedded in fat and connective tissue. The axillary lymph nodes are clinically paramount, particularly in the context of breast cancer, as they are often the first site of metastasis. Their assessment (palpation, imaging, biopsy) is crucial for staging and prognosis. Surgical procedures like axillary lymph node dissection (ALND) or sentinel lymph node biopsy (SLNB) are common, but carry risks of injury to adjacent structures, notably the long thoracic nerve (leading to winged scapula), thoracodorsal nerve, and intercostobrachial nerves (causing sensory disturbances). The brachial plexus itself is vulnerable during axillary surgery or due to trauma in this region. The axilla's crowded nature means that pathology in one structure can easily affect others, highlighting its significance as a high-stakes anatomical crossroads.

B. Cubital Fossa

The cubital fossa is a triangular depression on the anterior aspect of the elbow. Its boundaries are: superiorly, an imaginary line between the medial and lateral epicondyles of the humerus; medially, the pronator teres muscle; and laterally, the brachioradialis muscle. The floor is formed by the brachialis and supinator muscles, and the roof by skin, fascia, and the bicipital aponeurosis. The contents, from lateral to medial, are typically remembered by the mnemonic "Really Need Beer To Be At My Nicest" (though the radial nerve is often considered just lateral to or deep within the fossa rather than a direct content): Radial nerve (and its division into superficial and deep/posterior interosseous branches), Biceps brachii tendon, Brachial artery (which bifurcates into the radial and ulnar arteries at the apex of the fossa), and the Median nerve. Superficially within the roof lies the median cubital vein, which connects the cephalic and basilic veins. This vein is a

common site for venipuncture due to its accessibility and relative stability. The brachial artery pulse is palpable medial to the biceps tendon and is used for blood pressure measurement. The cubital fossa is clinically relevant in supracondylar fractures of the humerus, where displacement of fragments can injure the brachial artery and/or median nerve, potentially leading to serious complications like Volkmann's ischemic contracture. This superficial landmark, therefore, has deep clinical implications related to vascular access and neurovascular vulnerability.

C. Carpal Tunnel

The carpal tunnel is an osteofibrous passageway on the volar (palmar) aspect of the wrist. Its floor and walls are formed by the arch of the carpal bones, and its roof is formed by the strong, transverse flexor retinaculum (transverse carpal ligament). The flexor retinaculum attaches medially to the pisiform and hook of the hamate, and laterally to the scaphoid tubercle and trapezium. The carpal tunnel transmits nine flexor tendons (flexor pollicis longus, four flexor digitorum superficialis, and four flexor digitorum profundus) and the median nerve. The tendons are surrounded by synovial sheaths. The clinical significance of the carpal tunnel lies primarily in Carpal Tunnel Syndrome (CTS), the most common entrapment neuropathy. CTS occurs when the median nerve is compressed within this unyielding space, often due to inflammation or swelling of the tendon sheaths (tenosynovitis), trauma, anatomical variations, or systemic conditions like rheumatoid arthritis or pregnancy. The anatomy of the carpal tunnel, being a confined space with little room for expansion, directly dictates its propensity for median nerve compression when tissue volume within it increases. Symptoms include pain, paresthesia (tingling, numbness) in the median nerve distribution (lateral 3.5 digits), and thenar muscle weakness/atrophy in advanced cases.

D. Anatomical Snuffbox

The anatomical snuffbox is a triangular depression on the dorsolateral aspect of the wrist, best seen when the thumb is extended and abducted. Its boundaries are:

Anterolaterally (radial border): Tendons of the abductor pollicis longus (APL) and extensor pollicis brevis (EPB).

Posteromedially (ulnar border): Tendon of the extensor pollicis longus (EPL).

Floor (proximal to distal): Radial styloid process, scaphoid bone, trapezium bone, and base of the first metacarpal.

The contents of the anatomical snuffbox include the radial artery (as it courses towards the deep palmar arch), superficial branches of the radial nerve (providing cutaneous sensation), and the cephalic vein (originating here). The primary clinical

significance of the anatomical snuffbox is its utility in diagnosing scaphoid fractures. Tenderness on palpation within the snuffbox after a fall on an outstretched hand is highly suggestive of a scaphoid fracture, even if initial radiographs are negative. Given the risk of avascular necrosis and non-union associated with scaphoid fractures, this small anatomical triangle holds significant diagnostic value. The radial artery pulse can also be palpated in this location.

IV. Clinical Anatomy of Neurovascular Structures and Common Lesions

The intricate network of nerves and blood vessels in the upper limb is susceptible to injury from trauma, compression, or disease. Understanding their anatomical course and relationships is critical for diagnosing and managing these conditions.

A. Common Upper Limb Fractures and Associated Neurovascular Injuries

Fractures of the upper limb bones frequently place adjacent neurovascular structures at risk. The mechanism of injury, the degree of displacement, and the specific anatomical location of the fracture determine the likelihood and type of associated neurovascular compromise. These injuries are not merely skeletal disruptions but can serve as indicators of potential, and often more critical, damage to nerves and vessels.

Clavicle Fractures: Commonly occur from a fall onto the shoulder or direct trauma, typically at the junction of the middle and lateral thirds. Due to its proximity to the great vessels and brachial plexus as they pass between the clavicle and the first rib, displaced fractures or excessive callus formation can lead to compression or direct injury to the **subclavian artery/vein** or components of the **brachial plexus**. Delayed fixation also increases this risk due to scar tissue tethering these structures.

Proximal Humerus Fractures (especially Surgical Neck): Often occur in elderly individuals with osteoporosis or from high-energy trauma in younger patients. Fractures of the **surgical neck of the humerus** pose a significant risk to the **axillary nerve** and the **posterior circumflex humeral artery**, which wind around this part of the bone. Axillary nerve injury results in deltoid paralysis and sensory loss over the "regimental badge" area.

Supracondylar Humerus Fractures: These are common in children, usually resulting from a FOOSH causing hyperextension of the elbow. The distal fragment is often displaced posteriorly. The **brachial artery**, which lies anterior to the distal humerus, is vulnerable to laceration, kinking, or compression, potentially leading to Volkmann's ischemic contracture. The **median nerve**, which accompanies the brachial artery, is also frequently injured. The **radial**

nerve (if displacement is posteromedial) and **ulnar nerve** (especially in flexion-type fractures or with posterolateral displacement) can also be affected.

Distal Radius Fractures (e.g., Colles' Fracture): A Colles' fracture, characterized by dorsal displacement and angulation of the distal radial fragment, typically results from a FOOSH. Swelling and deformity at the fracture site can compress or stretch the **median nerve** as it passes through the carpal tunnel, leading to acute carpal tunnel syndrome.

Scaphoid Fractures: As previously discussed, the primary concern with scaphoid fractures is the risk of **avascular necrosis (AVN)** of the proximal fragment due to its retrograde blood supply from the radial artery. Non-union is another common complication, potentially leading to wrist arthritis.

B. Brachial Plexus and Peripheral Nerve Injuries

The brachial plexus, formed by the ventral rami of spinal nerves C5, C6, C7, C8, and T1, provides the entire motor and sensory innervation to the upper limb (with some exceptions like trapezius). Injuries to the plexus or its terminal branches can result in significant functional impairment.

Brachial Plexus Injury Patterns:

Erb's Palsy (Upper Trunk Lesion - C5, C6): Typically occurs due to excessive traction or stretching of the neck during difficult childbirth (shoulder dystocia) or from falls/trauma that forcefully separate the head and shoulder. It affects muscles innervated by C5 and C6, including the deltoid, supraspinatus, infraspinatus, biceps brachii, and brachialis. This results in the characteristic "waiter's tip" or "porter's tip" deformity: the arm hangs adducted and internally rotated, the elbow is extended, and the forearm is pronated, with the palm facing posteriorly or superiorly.

Klumpke's Palsy (Lower Trunk Lesion - C8, T1): Less common, typically caused by hyperabduction injuries (e.g., grabbing onto something while falling) or difficult breech delivery where the arm is pulled overhead. It affects the intrinsic muscles of the hand (innervated by T1) and flexors of the wrist and fingers (C8). This leads to a "claw hand" deformity (hyperextension at MCP joints and flexion at IP joints) and sensory loss along the medial aspect of the arm, forearm, and hand. Horner's syndrome (ptosis, miosis, anhydrosis) may be present if T1 sympathetic fibers are involved.

Pan-Plexus Injury (C5-T1): Results in a flail, anesthetic limb. These are often severe traction injuries.

Specific Peripheral Nerve Lesions:

Axillary Nerve (C5, C6 - from posterior cord) Anatomical Course:

Arises from the posterior cord in the axilla, passes through the quadrilateral space with the posterior circumflex humeral artery, and winds around the surgical neck of the humerus. Supplies deltoid and teres minor; provides sensation to the "regimental badge" area.

Sites of Vulnerability/Mechanisms:

Fracture of surgical neck, anterior dislocation of shoulder, crutch palsy.

Clinical Manifestations:

Deltoid paralysis (impaired arm abduction beyond 15°), teres minor weakness (external rotation), sensory loss over deltoid region.

Radial Nerve (C5–T1 - from posterior cord) Anatomical Course:

Descends in the radial groove of humerus with profunda brachii artery, divides into superficial (sensory) and deep (motor) branches at elbow.

Sites of Vulnerability/Mechanisms:

Humeral shaft fracture, prolonged compression ("Saturday night palsy"), crutch use.

Clinical Manifestations:

Wrist drop, inability to extend wrist and fingers, sensory loss on posterior arm, forearm, and dorsum of lateral hand.

Median Nerve (C5–T1 - from lateral and medial cords) Anatomical Course:

Formed in axilla, courses through cubital fossa, enters forearm between pronator teres heads, passes through carpal tunnel into hand.

Sites of Vulnerability/Mechanisms:

Supracondylar humerus fracture, pronator teres compression, carpal tunnel syndrome, wrist laceration.

Clinical Manifestations:

- *Elbow/proximal forearm lesion:* Weak pronation, wrist flexion, FDS/FDP paralysis, "Hand of Benediction" when making a fist, thenar atrophy.
- *Wrist lesion (CTS):* No forearm muscle weakness; sensory loss in lateral 3.5 digits, thenar wasting, "Ape hand" in chronic cases.

Ulnar Nerve (C8, T1 - from medial cord) Anatomical Course:

Travels posterior to medial epicondyle (cubital tunnel), enters forearm between FCU heads, passes through Guyon's canal into hand.

Sites of Vulnerability/Mechanisms:

Medial epicondyle fracture, cubital tunnel compression, Guyon's canal entrapment, wrist lacerations.

Clinical Manifestations:

- *Elbow lesion:* FCU and FDP (medial) weakness, sensory loss in medial 1.5 digits, intrinsic hand muscle paralysis, "ulnar claw," positive Froment's sign.
- *Wrist lesion (Guyon's canal):* Sparing of forearm muscles, variable sensory/motor hand deficits depending on site within canal.

The "Double-Crush" Phenomenon

A nerve that is already compromised at one location may be more susceptible to dysfunction if subjected to a secondary, even milder, insult elsewhere along its path. This "double-crush" phenomenon arises because the initial injury can impair axoplasmic flow, rendering the nerve less resilient. For example, a subclinical cervical radiculopathy affecting C6/C7 nerve roots might predispose the median nerve to develop more severe symptoms of carpal tunnel syndrome from a relatively mild compression at the wrist. This underscores the importance of a comprehensive neurological examination that assesses the entire course of an affected nerve, rather than focusing solely on the site of maximal symptoms, as proximal, seemingly unrelated issues could be contributing to or exacerbating the distal neuropathy.

C. Vascular Anatomy and Clinical Correlations

The vascular system of the upper limb ensures adequate perfusion for its diverse metabolic demands. Knowledge of arterial pathways, pulse points, and collateral circulation is fundamental for clinical assessment and procedures.

1. Course and Palpation of Major Arteries (Axillary, Brachial, Radial, Ulnar)

Axillary Artery: A continuation of the subclavian artery, it begins at the lateral border of the first rib, traverses the axilla, and becomes the brachial artery at the inferior border of the teres major muscle. It is anatomically divided into three parts by the pectoralis minor muscle. The axillary pulse can be palpated in the axilla, inferior to its lateral wall, by pressing the artery against the proximal humerus or lateral thoracic wall.

Brachial Artery: This is the continuation of the axillary artery, running down the medial aspect of the arm, initially medial to the humerus and then anteriorly to pass into the cubital fossa. In the cubital fossa, it lies medial to the biceps tendon and bifurcates into the radial and ulnar arteries. The brachial pulse is palpable here and is the site for auscultation during blood pressure measurement.

Radial Artery: Arising from the brachial artery bifurcation, it courses down the lateral (radial) side of the forearm, initially deep to the brachioradialis muscle. In the distal forearm, it becomes superficial, lying lateral to the tendon of the flexor carpi radialis (FCR). It then passes dorsally, through the anatomical snuffbox, to contribute significantly to the deep palmar arch. The radial pulse is most commonly palpated at the wrist, just lateral to the FCR tendon and proximal to the thumb. This site is frequently used for assessing heart rate and rhythm, and for arterial blood gas sampling or arterial line placement.

Ulnar Artery: The larger terminal branch of the brachial artery, it courses down the medial (ulnar) side of the forearm, initially deep to several forearm flexors, then running between the flexor carpi ulnaris (FCU) and flexor digitorum profundus (FDP). At the wrist, it passes superficial to the flexor retinaculum, lateral to the pisiform bone, through Guyon's canal, and is the main contributor to the superficial palmar arch. The ulnar pulse can be palpated at the wrist, just lateral to the FCU tendon and proximal to the pisiform bone.

2. The Allen's Test: Anatomical Basis and Clinical Application

The Allen's test is a clinical maneuver used to assess the patency of the radial and ulnar arteries and the integrity of the superficial and deep palmar arches, which provide crucial collateral circulation to the hand. The hand receives a dual blood supply from both the radial and ulnar arteries, which anastomose to form these arches. This redundancy is critical; if one artery is occluded, the other can often maintain perfusion.

Anatomical Basis: The test relies on the principle of collateral flow. By occluding both arteries and then releasing one, the examiner can observe the pattern and speed of reperfusion, indicating the adequacy of the released artery and the connecting arches.

Procedure: The examiner compresses both the radial and ulnar arteries firmly at the patient's wrist. The patient is asked to clench and unclench their hand several times to exsanguinate it, after which the hand is opened (but not hyperextended, to avoid a false positive) and should appear blanched. Pressure is then released from one artery (e.g., the ulnar artery), while maintaining pressure on the other (radial artery). The time taken for color (capillary refill) to return to the palm and fingers is observed. A normal refill time is typically considered less than 5 to 7 seconds. The test is then repeated, releasing the other artery.

Clinical Application: The Allen's test is most commonly performed before procedures that might compromise one of these arteries, such as radial artery cannulation for arterial blood gas analysis or continuous blood pressure monitoring, or prior to harvesting the radial artery for coronary artery bypass grafting (CABG). A positive test (rapid refill) suggests adequate collateral circulation from the non-occluded artery, making the procedure safer. A negative or equivocal test may prompt further investigation or selection of an alternative site.

3. Vascular Compromise in Upper Limb Trauma

Arterial injuries in the upper limb are serious and can be limb-threatening. They can result from lacerations, contusions leading to thrombosis or spasm, or compression by displaced bone fragments or hematoma.

Mechanisms: As discussed previously, supracondylar fractures of the humerus are notorious for risking brachial artery injury. Elbow dislocations can also compromise the brachial artery. Shoulder dislocations may injure the axillary artery, and clavicular fractures can, albeit less commonly, affect the subclavian artery. Penetrating trauma can damage any vessel along its path.

Signs of Acute Arterial Occlusion: The classic signs are often remembered by the "6 Ps": **P**ain (often severe and out of proportion), **P**allor (pale skin), **P**ulselessness (distal to the injury), **P**aresthesia (tingling, numbness), **P**aralysis (weakness or inability to move), and **P**oikilothermia (coolness of the limb compared to the other side). Prompt recognition and intervention (e.g., reduction of fracture/dislocation, vascular repair) are essential to restore perfusion and prevent irreversible tissue damage.

Compartment Syndrome: This is a critical complication, especially following forearm fractures, supracondylar fractures, or crush injuries. Increased pressure within a closed fascial compartment compromises circulation and nerve function within that compartment. It is characterized by severe pain (especially with passive stretch of muscles in the compartment), tenseness of the compartment, paresthesia, and later, paralysis and pulselessness (a late sign). Urgent fasciotomy is required to decompress the compartment and prevent permanent damage.

While the palmar arches generally provide robust collateral circulation to the hand, significant anatomical variation exists in their completeness and the relative dominance of the radial versus ulnar artery contributions. The Allen's test, being a subjective visual assessment of capillary refill, may not always perfectly predict the hand's ability to tolerate occlusion of one artery, particularly if one artery is hypoplastic or an arch is incomplete. Factors such as skin pigmentation, ambient temperature, and observer experience can influence its interpretation. This inherent variability and the test's limitations highlight why, in high-risk scenarios or when the test results are ambiguous, more objective assessments like Doppler ultrasonography are increasingly considered to evaluate vascular patency and collateral flow more definitively. This illustrates a critical point in clinical practice: standard tests are valuable, but their interpretation must always be contextualized by an understanding of potential anatomical variations and the availability of more precise diagnostic tools when needed.

Table 1: Key Bony Features and Articulations of Upper Limb Bones

Bone	Key Features	Proximal Articulation(s)	Distal Articulation(s)	Common Fracture Sites/Clinical Notes
Clavicle	Sternal end, acromial end, conoid tubercle, trapezoid line	Manubrium (sternoclavicular joint)	Acromion of scapula (acromioclavicular joint)	Junction of middle and lateral thirds. Risk to subclavian vessels/brachial plexus.
Scapula	Glenoid cavity, acromion, coracoid process, spine, supraspinous/ infraspinous fossae	Clavicle (acromioclavicular joint)	Humerus (glenohumeral joint)	Fractures less common, usually due to high-energy trauma. Winging of scapula with long thoracic nerve injury.
Humerus	Head, anatomical/surgical necks, greater/lesser tubercles, intertubercular groove, deltoid tuberosity, radial groove, medial/lateral epicondyles, trochlea, capitulum, olecranon/coronoid/radial fossae	Scapula (glenohumeral joint)	Radius (capitulum joint), Ulna (trochleo-ulnar joint)	Surgical neck (axillary nerve risk), shaft (radial nerve risk), supracondylar region (brachial artery, median nerve risk).
Radius	Head, neck, radial tuberosity, styloid process, ulnar notch	Humerus (capitulum), Ulna (proximal radioulnar joint)	Scaphoid, Lunate (radiocarpal joint), Ulna (distal radioulnar joint)	Distal radius (Colles' fracture, Smith's fracture), radial head.
Ulna	Olecranon process, coronoid process, trochlear notch, radial notch, head, styloid process	Humerus (trochlea)	Radius (distal radioulnar joint), Articular disc of wrist	Olecranon, ulnar shaft ('nightstick' fracture).
Carpals	Proximal row: Scaphoid, Lunate, Triquetrum, Pisiform. Distal row: Trapezium, Trapezoid, Capitate, Hamate.	Radius, articular disc (proximal row)	Metacarpals (distal row)	Scaphoid (avascular necrosis risk). Lunate (dislocation, Kienböck's disease).
Metacarpals (I-V)	Base, shaft, head	Distal carpal row, adjacent metacarpals	Proximal phalanges (metacarpophalangeal joints)	Neck of 5th metacarpal (Boxer's fracture), base of 1st metacarpal (Bennett's, Rolando fractures).
Phalanges	Proximal, Middle (II-V), Distal. Each has base, shaft, head.	Metacarpals (proximal), Proximal phalanges (middle/distal)	Middle phalanges (distal in II-V), Interphalangeal joints	Tuft fractures, avulsion fractures (e.g., Mallet finger).

Table 2: Common Upper Limb Fractures and Associated Neurovascular Complications

Fracture Site	Common Mechanism(s)	Primary Neurovascular Structures at Risk	Potential Clinical Sequelae
Clavicle (Mid-shaft/Lateral)	Fall on shoulder, direct blow	Subclavian artery/vein, Brachial plexus (trunks/divisions)	Vascular compromise, thoracic outlet syndrome, brachial plexopathy
Humerus (Surgical Neck)	FOOSH, direct blow to shoulder	Axillary nerve, Posterior circumflex humeral artery	Deltoid paralysis, sensory loss (regimental badge), hematoma
Humerus (Shaft)	Direct blow, fall, twisting injury	Radial nerve (in radial groove), Profunda brachii artery	Wrist drop, sensory loss (dorsal hand), impaired elbow extension (if high)
Humerus (Supracondylar)	FOOSH (hyperextension)	Brachial artery, Median nerve, Radial nerve, Ulnar nerve	Volkman's ischemic contracture, median/radial/ulnar nerve palsies, compartment syndrome
Elbow (Dislocation)	FOOSH	Brachial artery, Ulnar nerve, Median nerve	Vascular compromise, nerve palsies, compartment syndrome
Radius (Distal - Colles')	FOOSH (wrist extension)	Median nerve	Acute carpal tunnel syndrome, paresthesia in median nerve distribution
Scaphoid	FOOSH (wrist extension/radial deviation)	Radial artery (branches to scaphoid)	Avascular necrosis of proximal pole, non-union, chronic wrist pain

Table 3: Major Nerves of the Upper Limb: Course, Motor/Sensory Function, and Common Injury/Entrapment Syndromes

Nerve	Root Value(s)	General Course	High-lights	Key Muscles Innervated	Main Cutaneous Innervation Area	Common Sites of Injury/Entrapment	Characteristic Clinical Presentation of Lesion
Axillary N.	C5, C6	Exits axilla via quadrilateral space, winds around surgical neck of humerus.		Deltoid, Teres minor.	Skin over lower deltoid ("regimental badge").	Surgical neck of humerus fracture, anterior shoulder dislocation, crutch palsy.	Inability to abduct arm (beyond 15°), weak external rotation, deltoid atrophy, sensory loss over badge area.
Radial N.	C5-T1	Passes in radial groove of humerus, divides in cubital fossa/supinator.		Triceps brachii, Brachioradialis, Supinator, Extensors of wrist & digits.	Posterior arm, posterior forearm, dorsal aspect of lateral 3.5 digits (variable).	Humeral shaft fracture (spiral groove), "Saturday night palsy" (compression in axilla/spiral groove), crutch palsy.	Wrist drop, finger drop, weakness of supination, sensory loss. Triceps spared if lesion distal to its innervation.
Median N.	C5-T1	Descends with brachial artery, passes through cubital fossa, between heads of pronator teres, through carpal tunnel.		Forearm flexors (except FCU & ulnar FDP), Thenar muscles, lateral 2 Lumbricals.	Palmar aspect of lateral 3.5 digits & dorsal tips.	Supracondylar humerus fracture, Pronator teres syndrome, Carpal Tunnel Syndrome (CTS), wrist lacerations.	"Ape hand" (thenar atrophy, loss of opposition - late), "Hand of Benediction" (proximal lesion, on attempting fist), sensory loss. CTS: nocturnal paresthesia, thenar weakness.
Ulnar N.	C8, T1	Passes posterior to medial epicondyle (cubital tunnel), enters forearm between heads of FCU, passes through Guyon's canal at wrist.		Flexor carpi ulnaris (FCU), ulnar half of Flexor digitorum profundus (FDP), Hypothenar muscles, Interossei, medial 2 Lumbricals, Adductor pollicis.	Medial 1.5 digits (palmar & dorsal).	Medial epicondyle fracture/trauma, Cubital Tunnel Syndrome, Guyon's canal compression, wrist lacerations.	"Ulnar claw hand" (4th & 5th digits), Froment's sign (weak thumb adduction), wasting of intrinsic hand muscles, sensory loss in ulnar distribution.
Musculocut. N.	C5, C6, C7	Pierces coracobrachialis, runs between biceps & brachialis, becomes lateral cutaneous nerve of forearm.		Coracobrachialis, Biceps brachii, Brachialis.	Lateral forearm.	Rare in isolation; penetrating trauma, shoulder dislocation.	Weak elbow flexion (biceps, brachialis), weak supination (biceps), sensory loss on lateral forearm.